Network analysis of marine food webs from a species-centric approach

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I. INTRODUCTION

Marine environments constitute the majority of Earth's surface, perform vital ecosystem services, and serve as a food source for humans and other organisms. Safeguarding marine ecosystems involves analyzing their health to create informed, functional policies for appropriate, sustained human use. One indicator of ecosystem health is the persistence of a diverse, robust food web. Food webs employ feeding relationships to track how energy travels throughout an ecosystem. Our goal for this project is to evaluate feeding relationships among marine organisms to construct a food web network and utilize network characteristics to draw conclusions about ecosystem health and resilience. We tackle the following questions: How does community composition shape the creation and persistence of a marine ecosystem's food web? Are species higher up in the food chain more connected and/or exhibit greater diet diversity than primary producers and low-level consumers? How robust is a food web to perturbations? We hypothesize that a range of primary producers, primary consumers, secondary consumers, tertiary consumers, and decomposers are required to create a robust food web within a marine ecosystem, and we hypothesize upper-level consumers will be highly connected species and exhibit greater diet diversity compared to primary producers and low-level consumers because they feed on a range of energy sources.

The Gulf of Cadiz is a highly productive and biodiverse marine ecosystem that has previously been examined for its biological and ecological characteristics of the region. Previous studies have used ecosystem and mass-balance models to characterize trophic interactions and energy flows. Torres et al. [1] modeled 43 functional groups and identified major trophic pathways and keystone groups, while Llope [2] emphasized the importance of ecosystem-based management in the region. More broadly, studies of marine food webs have shown that community composition influences network structure [3] and that species can differ substantially in their network positions based on ecological characteristics [4]. As a result, network measures, such as degree, centrality, connectance, and trophic level, have become useful for describing food web organization and identifying ecologically important species [5, 6].

Marine food webs have also been studied as complex networks that can provide insight into ecosystem stability and health. Studies have shown that diversity and connectance can influence food-web persistence and robustness [7, 8] while network topology and species interactions can be used to determine ecosystem health [9]. Recent research has also shown that the relationship between food web structure and ecosystem resilience is seen across global marine ecosystems [10]. Although the Gulf of Cadiz has been studied in terms of trophic flows, ecosystem functioning, and fishing impacts, not as much work has been done to understand network properties of individual species.

Despite the extensive research conducted on the Gulf of Cadiz, there is room for further study in the way of understanding the relationship between individual species' trophic levels and their diet diversity. Current research focuses on characterizing ecosystem-level trophic interactions and food web structure. We aim to build upon this foundation by synthesizing information regarding the relative importance of prey in each predator's diet and an estimation of their trophic levels. Our project intends to build on the existing Gulf of Cadiz food web model by examining which species are most connected and whether their connectivity and diet diversity vary with trophic level, helping us understand how community composition shapes food web structure and potential ecosystem resilience. Considering the rapid rate of environmental change driven by anthropogenic activity, developing the network tools to evaluate relative importance of individual predator and prey species in establishing feeding relationships within marine ecosystems, which serve as biodiversity hotspots, is indispensable.

DATA AVAILABILITY

All code and data used to generate the results in this study are available on GitHub.

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